Propositional Logic

Proposition

A proposition is a **declarative statement**. It must have a definite truth value, which means it is **either true or false, but not both at the same time**. In logic, propositions are usually represented in **logical or mathematical form**.

Examples of propositions

Some examples written on the board are:

1. **It is Sunday** → may be **True/False** depending on the day
2. **The sun rises from the west** → **False**
3. **3 + 3 = 7** → **False**
4. **5 is a prime number** → **True**

Truth values and Boolean logic

In propositional logic, truth values are often written as:

- **T → 1**
- **F → 0**

This is called **Boolean Logic (0/1)**.

What is not a proposition

The board also notes that the following are **not propositions** because they do not have fixed truth values in the logical sense:

- **Questions**
- **Commands**
- **Opinions**

Types of propositions

Propositions can be of two types:

a) Atomic proposition

An **atomic proposition** is a simple statement that cannot be broken into smaller logical parts.

Example:
2 + 2 = 4 → True

b) Compound proposition

A **compound proposition** is formed by combining two or more propositions using logical connectives.

Example:
Let,
P = It is raining
Q = The street is wet

Then a compound proposition is:
P and Q = *It is raining and the street is wet*

Logical Connectives (Operators)

There are five basic logical connectives shown in the board content.

1. Negation

Symbol: ¬
Meaning: **not**

If
P = Today is Friday

then
¬P = Today is not Friday

Negation simply reverses the truth value of a proposition.

2. Conjunction

Symbol: ∧
Meaning: and

If
P = Rohan is intelligent
Q = Rohan is hardworking

then
P ∧ Q = Rohan is intelligent and hardworking

Conjunction is true only when both propositions are true.

3. Disjunction

Symbol: ∨
Meaning: or

If
P = Ritika is a doctor
Q = Ritika is an engineer

then
P ∨ Q = Ritika is a doctor or an engineer

Disjunction means at least one of the statements is true.

4. Implication

Symbol: →
Meaning: if ... then

If
P = It is raining
Q = The street is wet

then
P → Q = If it is raining, then the street is wet

This shows a condition and its result.

5. Biconditional

Symbol: ↔
Meaning: iff or if and only if

If
P = I will go to mall
Q = I have to do shopping

then
P ↔ Q = I will go to mall if and only if I have to do shopping

A biconditional means both statements imply each other.

Truth Table of Basic Logical Connectives

P	Q	P ∧ Q	P ∨ Q	P → Q	P ↔ Q
T	T	T	T	T	T
T	F	F	T	F	F
F	T	F	T	T	F
F	F	F	F	T	T

Conjunction:

P ∧ Q means P and Q.
It is true only when both P and Q are true.

Disjunction:

P V Q means P or Q.
It is true when at least one of P or Q is true.

Implication:

P → Q means if P then Q.
It is false only when P is true but Q is false.

Biconditional:

P ↔ Q means P if and only if Q.
It is true when both propositions have the same truth value.

Important Logical Equivalence

One important equivalence written on the board is:

A → B = ¬A V B

This means that an implication can be rewritten as "not A or B". "If A then B" is logically equivalent to "not A or B".

Truth table to prove the equivalence

A	B	A → B	¬A	¬A V B
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Conclusion

The columns of A → B and ¬A V B are exactly the same. So,

$A \rightarrow B \equiv \neg A \vee B$

That is why they are logically equivalent.

Properties of Operators

Commutativity

The order of propositions can be changed without affecting the result in some operators.

Example:

$P \wedge Q = Q \wedge P$

$P \vee Q = Q \vee P$

Associativity

Grouping of propositions does not affect the result in some cases.

Example:

$(P \wedge Q) \wedge R = P \wedge (Q \wedge R)$

$(P \vee Q) \vee R = P \vee (Q \vee R)$

Identity Element

There are special truth values that keep a proposition unchanged.

Example:

$P \wedge T = P$

$P \vee F = P$

Distributive Law

One operator can be distributed over another.

Example:

$P \wedge (Q \vee R) = (P \wedge Q) \vee (P \wedge R)$

De Morgan's Law

Used to simplify negations of compound propositions.

Example:

$\neg(P \wedge Q) = \neg P \vee \neg Q$
$\neg(P \vee Q) = \neg P \wedge \neg Q$

Double Negation Elimination

If a proposition is negated twice, it returns to the original form.
Example:
$\neg(\neg P) = P$

Evaluating a Compound Logical Statement

Question

Given statement, "If the earth moves round the sun or the sun moves round the earth, then Copernicus might be a mathematician but wasn't an astronomer."

Find whether the meaning of the statement is true or false.

1. Define the simple propositions

Let,

P = The earth moves round the sun = T
Q = The sun moves round the earth = F
R = Copernicus might be a mathematician = T
S = Copernicus was an astronomer = T

2. Convert the statement into symbolic form

The statement is:

$(P \vee Q) \rightarrow (R \wedge \neg S)$

Explanation:

- "the earth moves round the sun **or** the sun moves round the earth" → **P ∨ Q**
- "Copernicus might be a mathematician **but** wasn't an astronomer" → **R ∧ ¬S**

So the full statement becomes:

(P ∨ Q) → (R ∧ ¬S)

3. Substitute the truth values

We know:

- P = T
- Q = F
- R = T
- S = T

So,

(P ∨ Q) → (R ∧ ¬S)
= **(T ∨ F) → (T ∧ ¬T)**

Step 4: Simplify

$$(\boldsymbol{T} \vee \boldsymbol{F}) = \boldsymbol{T}$$

$$\neg \boldsymbol{T} = \boldsymbol{F}$$

$$(\boldsymbol{T} \wedge \boldsymbol{F}) = \boldsymbol{F}$$

So,

$$\boldsymbol{T} \rightarrow \boldsymbol{F}$$

Using implication rule,

$$\boldsymbol{A} \rightarrow \boldsymbol{B} \equiv \neg \boldsymbol{A} \vee \boldsymbol{B}$$

Therefore,

$$\boldsymbol{T} \rightarrow \boldsymbol{F} \equiv \neg \boldsymbol{T} \vee \boldsymbol{F}$$
$$= \boldsymbol{F} \vee \boldsymbol{F}$$
$$= \boldsymbol{F}$$

Final Answer

$$(\boldsymbol{P} \vee \boldsymbol{Q}) \rightarrow (\boldsymbol{R} \wedge \neg \boldsymbol{S}) = \boldsymbol{F}$$

Therefore, the given statement is false.

<u>Question</u>

"Inspite of having French nationality, B. Russel was a critic of Imperialism, then either he was not a bachelor or he was a universal lover."

Step 1: Assign propositions

P = B. Russel has French nationality = T
Q = He was a critic of Imperialism = T
R = He was a bachelor = T
S = He was a universal lover = T

Step 2: Symbolic form

(P ∧ Q) → (¬R ∨ S)

(P ∧ Q) → (¬R ∨ S)
= (T ∧ T) → (¬T ∨ T)
= T → (F ∨ T)
= T → T
= ¬T ∨ T

$= F \vee T$
$= T$

Question

Prove that

$\neg(P \vee (\neg P \wedge Q)) \equiv \neg P \wedge \neg Q \equiv \neg(P \vee Q)$

Proof

Let,

LHS $= \neg(P \vee (\neg P \wedge Q))$
$= \neg P \wedge \neg(\neg P \wedge Q)$; [De Morgan's Law]
$= \neg P \wedge (P \vee \neg Q)$; [Double Negation and De Morgan's Law]
$= (\neg P \wedge P) \vee (\neg P \wedge \neg Q)$; [Distributive Law]
$= F \vee (\neg P \wedge \neg Q)$
$= \neg P \wedge \neg Q$
$= \neg(P \vee Q)$; [De Morgan's Law]

Hence, proved.

Rules of Inference

For a conditional statement, let

1. <u>Implication: $P \rightarrow Q$</u>

From this implication, three related forms are obtained.

2. Converse

$Q \rightarrow P$

The converse is formed by interchanging the antecedent and the consequent.

3. Contrapositive

$\neg Q \rightarrow \neg P$

The contrapositive is formed by negating both parts of the implication and then interchanging them.

4. Inverse

$\neg P \rightarrow \neg Q$

The inverse is formed by negating both the antecedent and the consequent.

Truth Table

P	Q	¬P	¬Q	P → Q	Q → P	¬Q → ¬P	¬P → ¬Q
0	0	1	1	1	1	1	1
0	1	1	0	1	0	1	0
1	0	0	1	0	1	0	1
1	1	0	0	1	1	1	1

Here,
1 = True
0 = False

From the truth table, we get:

Implication = Contrapositive
That means,

$P \rightarrow Q \equiv \neg Q \rightarrow \neg P$

Also,

Converse = Inverse
That means,

$Q \rightarrow P \equiv \neg P \rightarrow \neg Q$

Types of Inference Rules

1. Modus Ponens

Modus Ponens is a valid rule of inference. Its form is:

$$P \rightarrow Q$$
$$P$$
$$\therefore Q$$

This means, if a conditional statement is true and the first part of that statement is also true, then the second part must be true.

Example

Statement 1: If I am sleepy, then I go to bed.
So, $P \rightarrow Q$

Here,
P = I am sleepy
Q = I go to bed

Statement 2: I am sleepy.
So, **P**

Conclusion: I go to bed.
So, **Q**

Therefore, from **P → Q** and **P**, we can conclude **Q**.

Main idea

Modus Ponens follows the pattern:

If P then Q
P is true
Therefore, **Q is true**

To check validity, we look at the rows where both premises are true, that is, where P → Q is true and P is true.

P	Q	P → Q	Premise P	Conclusion Q
0	0	1	0	0
0	1	1	0	1
1	0	0	1	0
1	1	1	1	1

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2. Modus Tollens

Modus Tollens is another valid rule of inference. Its form is:

$$P \rightarrow Q$$
$$\neg Q$$
$$\therefore \neg P$$

This means, if a conditional statement is true and the second part is false, then the first part must also be false.

Example

Statement 1: If I am sleepy, then I go to bed.
So, **P → Q**

Here,
P = I am sleepy
Q = I go to bed

Statement 2: I do not go to bed.
So, **¬Q**

Conclusion: I am not sleepy.
So, **¬P**

Therefore, from **P → Q** and **¬Q**, we can conclude **¬P**.

Main idea

Modus Tollens follows the pattern:

If P then Q
Q is false
Therefore, **P is false**

To check validity, we look at the rows where both premises are true, that is, where P → Q is true and ¬Q is true.

P	Q	¬P	¬Q	P → Q	Premise ¬Q	Conclusion ¬P
0	0	1	1	1	1	1
0	1	1	0	1	0	1
1	0	0	1	0	1	0
1	1	0	0	1	0	0

Predicate Logic

Predicate Logic is used when we need to express statements involving objects, properties, and relations more precisely than propositional logic. It is especially useful for statements such as:

1. Some humans are intelligent.
2. All birds can fly.

Predicate Logic is also known as First-Order Logic.

Main parts of Predicate Logic

Predicate Logic mainly involves two important aspects:

Syntax

Syntax of Predicate Logic determines which collection of symbols forms a logical expression.

Semantics

Semantics explains the meaning or interpretation of those logical expressions.

Basic symbols used in Predicate Logic

Symbol Type	Representation
Constant	1, 2, Asad, Dhaka, Cat
Variable	x, y, z, a, b
Predicate / Relation	Father, Brother
Function	sqrt
Connectives	$\wedge$, $\vee$, $\neg$, $\rightarrow$, $\leftrightarrow$
Equality	$=$, $\neq$
Quantifier	$\forall$, $\exists$

Atomic Sentence

An atomic sentence is the most basic sentence in predicate logic.

General form:

Predicate(term1, term2, term3, ...)

Examples

1. Karan and Arjun are brothers
   brothers(Karan, Arjun)
2. Chinky is a cat
   Cat(Chinky)

Here, “brothers” shows a relation between two terms, while “Cat” shows a property of one term.

Complex Sentence

A complex sentence is a combination of atomic sentences.

That means when two or more atomic sentences are connected using logical connectives, they form a complex sentence.

Quantifiers

Quantifiers are used in predicate logic to express how many objects satisfy a given property.

They are mainly of two types:

1. Universal Quantifier
2. Existential Quantifier

1. Universal Quantifier

The universal quantifier is used for statements such as:

- for all

- for each
- for every

Symbol used:

∀

Example:

All boys like cricket.

Predicate form:

∀x (Boy(x) → Like(x, Cricket))

This means:
For every x, if x is a boy, then x likes cricket.

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2. Existential Quantifier

The existential quantifier is used for statements such as:

- there exists
- for some
- for at least one

Symbol used:

∃

Example:

Some boys like cricket.

Predicate form:

∃x (Boy(x) ∧ Like(x, Cricket))

This means:
There exists at least one x such that x is a boy and x likes cricket.

Common predicate used in the example

Like(Boys, Cricket)

More properly in predicate logic, we write:

Like(x, Cricket)

where x represents a variable.

Type	Meaning	Symbol	Example sentence	Predicate form
Universal Quantifier	For all, for each, for every	∀	All boys like cricket	∀x (Boy(x) → Like(x, Cricket))
Existential Quantifier	There exists, for some, for at least one	∃	Some boys like cricket	∃x (Boy(x) ∧ Like(x, Cricket))

Properties of Quantifiers

1. ∀x ∀y = ∀y ∀x

This means the order of two universal quantifiers can be changed without affecting the meaning.

2. ∃x ∃y = ∃y ∃x

This means the order of two existential quantifiers can also be changed without affecting the meaning.

3. ∀x ∃y ≠ ∃y ∀x

This means the order cannot generally be changed when one is a universal quantifier and the other is an existential quantifier.

So, in short:

- Universal with universal: order can change
- Existential with existential: order can change
- Universal with existential: order usually cannot change

Quantifier Duality

Quantifier duality shows the relation between universal and existential quantifiers through negation.

1. All boys like cricket

$\forall x$ Like(x, Cricket) is equivalent to $\neg\exists x$ ¬Like(x, Cricket)

Meaning:
"All boys like cricket" means "there does not exist any boy who does not like cricket."

2. Some boys like chocolate

$\exists x$ Like(x, Chocolate) is equivalent to $\neg\forall x$ ¬Like(x, Chocolate)

Meaning:
"Some boys like chocolate" means "it is not true that all boys do not like chocolate."

General duality rules

$\forall x\ P(x) \equiv \neg\exists x\ \neg P(x)$

$\exists x\ P(x) \equiv \neg\forall x\ \neg P(x)$

Question: Inference from the given knowledge

Given statements

(a) If it is hot and humid, then it is raining.
(b) If it is humid, then it is hot.
(c) It is humid.

Is it possible to infer, “It is raining”?

Soln:

Let,

P = It is hot = T
Q = It is humid = T
R = It is raining = ?

(a) $(P \land Q) \rightarrow R$
(b) $Q \rightarrow P$
(c) Q

From (b),

$Q \rightarrow P$
Since Q = T, therefore P = T

Now from (a),

$(P \land Q) \rightarrow R$

Using, $A \rightarrow B = \neg A \lor B$

So,

$(P \land Q) \rightarrow R$
$= \neg(P \land Q) \lor R$
$= (\neg P \lor \neg Q) \lor R$

$= (\neg T \vee \neg T) \vee R$
$= (F \vee F) \vee R$
$= F \vee R$
$= R$

Therefore,

R = It is raining.

Conclusion:

Yes, it is possible to infer “It is raining”.

Question:

Given Knowledge:

(a) If you have a current password, then you can log on to the network.
(b) You have a current password.

Infer: You can log on to the network.

Soln:

Let,

P = You have a current password
Q = You can log on to the network

(a) $P \rightarrow Q$
(b) P

From (a),

$P \rightarrow Q$
$= \neg P \vee Q$
$= \neg T \vee Q$

= F ∨ Q
= Q

Therefore,

Q = You can log on to the network.

Conclusion:

You can log on to the network.

Question:

Given,

(i) Every student is sincere
(ii) All who are sincere and hard worker will succeed in their career
(iii) Meena is hard worker
(iv) Meena is student

Will Meena succeed in her career?

Soln:

(i) ∀x [Student(x) → Sincere(x)]
(ii) ∀x [(Sincere(x) ∧ Hard_worker(x)) → Succeed_career(x)]
(iii) Hard_worker(Meena)
(iv) Student(Meena)

From eq (i),

Student(Meena) → Sincere(Meena)
⇒ ¬Student(Meena) ∨ Sincere(Meena)
⇒ ¬T ∨ Sincere(Meena)

⇒ F ∨ Sincere(Meena)
⇒ Sincere(Meena)

Now from eq (ii),

Sincere(Meena) ∧ Hard_worker(Meena) → Succeed_career(Meena)
⇒ (T ∧ T) → Succeed_career(Meena)
⇒ T → Succeed_career(Meena)
⇒ ¬T ∨ Succeed_career(Meena)
⇒ F ∨ Succeed_career(Meena)
⇒ Succeed_career(Meena)

Therefore, Meena will succeed in her career.

Question:

1. Consider the following axioms:

   a) Arif, Leon, and Sohan are players in coach Nadeem's academy.
   b) All players in coach Nadeem's academy train under Nadeem.
   c) Nadeem is a coach.
   d) Anyone who trains under a coach either respects or resents that coach.
   e) If a player respects a coach, then the player follows that coach's training plan.
   f) If a player resents a coach, then the player does not follow that coach's training plan.
   g) A player submits a protest only against a coach whose training plan the player does not follow.
   h) Anyone who refuses to attend a coach's fitness session resents that coach.
   i) Arif submitted a protest against Nadeem.
   j) Sohan refused to attend Nadeem's fitness session.

k) Leon attended every fitness session arranged by Nadeem and never submitted a protest against him.
l) Anyone who attended every fitness session of a coach and never submitted a protest against that coach respects that coach.

Determine whether Arif, Leon, and Sohan follow Nadeem's training plan. Show the logical derivation clearly.

Solution

We need to determine whether **Arif**, **Leon**, and **Sohan** follow **coach Nadeem's training plan**.

Step 1: Define predicates

Let,

- **Player(x)** = x is a player in Nadeem's academy
- **Train(x, y)** = x trains under y
- **Coach(y)** = y is a coach
- **Respect(x, y)** = x respects y
- **Resent(x, y)** = x resents y
- **Follow(x, y)** = x follows y's training plan
- **Protest(x, y)** = x submitted a protest against y
- **Refuse(x, y)** = x refused to attend y's fitness session
- **AttendAll(x, y)** = x attended every fitness session arranged by y
- **NeverProtest(x, y)** = x never submitted a protest against y

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Step 2: Write the axioms in symbolic form

From the given knowledge:

(a) Player(Arif), Player(Leon), Player(Sohan)
(b) $\forall x$ [Player(x) $\rightarrow$ Train(x, Nadeem)]
(c) Coach(Nadeem)
(d) $\forall x$ [(Train(x, Nadeem) $\wedge$ Coach(Nadeem)) $\rightarrow$ (Respect(x, Nadeem) $\vee$ Resent(x, Nadeem))]

(e) ∀x [(Player(x) ∧ Respect(x, Nadeem)) → Follow(x, Nadeem)]
(f) ∀x [(Player(x) ∧ Resent(x, Nadeem)) → ¬Follow(x, Nadeem)]
(g) ∀x [Protest(x, Nadeem) → ¬Follow(x, Nadeem)]
(h) ∀x [Refuse(x, Nadeem) → Resent(x, Nadeem)]
(i) Protest(Arif, Nadeem)
(j) Refuse(Sohan, Nadeem)
(k) AttendAll(Leon, Nadeem) ∧ NeverProtest(Leon, Nadeem)
(l) ∀x [(AttendAll(x, Nadeem) ∧ NeverProtest(x, Nadeem)) → Respect(x, Nadeem)]

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1. For Arif

From **(i)**,

Protest(Arif, Nadeem)

From **(g)**,

Protest(Arif, Nadeem) → ¬Follow(Arif, Nadeem)

Using **A → B = ¬A ∨ B**,

Protest(Arif, Nadeem) → ¬Follow(Arif, Nadeem)
= **¬Protest(Arif, Nadeem) ∨ ¬Follow(Arif, Nadeem)**
= **¬T ∨ ¬Follow(Arif, Nadeem)**
= **F ∨ ¬Follow(Arif, Nadeem)**
= **¬Follow(Arif, Nadeem)**

So,

Arif does not follow Nadeem's training plan.

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2. For Sohan

From **(j)**,

Refuse(Sohan, Nadeem)

From (h),

Refuse(Sohan, Nadeem) → Resent(Sohan, Nadeem)

Using A → B = ¬A ∨ B,

Refuse(Sohan, Nadeem) → Resent(Sohan, Nadeem)
= ¬Refuse(Sohan, Nadeem) ∨ Resent(Sohan, Nadeem)
= ¬T ∨ Resent(Sohan, Nadeem)
= F ∨ Resent(Sohan, Nadeem)
= Resent(Sohan, Nadeem)

Now from (a),

Player(Sohan)

From (f),

(Player(Sohan) ∧ Resent(Sohan, Nadeem)) → ¬Follow(Sohan, Nadeem)

Since Player(Sohan) = T and Resent(Sohan, Nadeem) = T,

(T ∧ T) → ¬Follow(Sohan, Nadeem)
= T → ¬Follow(Sohan, Nadeem)
= ¬T ∨ ¬Follow(Sohan, Nadeem)
= F ∨ ¬Follow(Sohan, Nadeem)
= ¬Follow(Sohan, Nadeem)

So,

Sohan does not follow Nadeem's training plan.

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3. For Leon

From (k),

AttendAll(Leon, Nadeem) ∧ NeverProtest(Leon, Nadeem)

From (l),

(AttendAll(Leon, Nadeem) ∧ NeverProtest(Leon, Nadeem)) → Respect(Leon, Nadeem)

So,

(T ∧ T) → Respect(Leon, Nadeem)
= T → Respect(Leon, Nadeem)
= ¬T ∨ Respect(Leon, Nadeem)
= F ∨ Respect(Leon, Nadeem)
= Respect(Leon, Nadeem)

Now from (a),

Player(Leon)

From (e),

(Player(Leon) ∧ Respect(Leon, Nadeem)) → Follow(Leon, Nadeem)

Since Player(Leon) = T and Respect(Leon, Nadeem) = T,

(T ∧ T) → Follow(Leon, Nadeem)
= T → Follow(Leon, Nadeem)
= ¬T ∨ Follow(Leon, Nadeem)
= F ∨ Follow(Leon, Nadeem)
= Follow(Leon, Nadeem)

So,

Leon follows Nadeem's training plan.

Final Conclusion

- Arif does not follow Nadeem's training plan.
- Leon follows Nadeem's training plan.
- Sohan does not follow Nadeem's training plan.

2. Consider the following axioms:
   a) Anyone who has a valid ID card and has completed safety training may enter the secure lab.
   b) Anyone whose access is suspended may not enter the secure lab.
   c) Anyone who missed the annual compliance review has suspended access.
   d) Anyone who submitted all compliance documents did not miss the annual compliance review.
   e) Nabila has a valid ID card.
   f) Nabila has completed safety training.
   g) Rafi has a valid ID card.
   h) Rafi has completed safety training.
   i) Rafi missed the annual compliance review.
   j) Toma has a valid ID card.
   k) Toma has completed safety training.
   l) Toma submitted all compliance documents.

Determine whether Nabila, Rafi, and Toma may enter the secure lab. Show the logical derivation clearly.

3. Consider the following axioms:
   a) Any patient with bacterial infection is prescribed Antibiotic-A unless the patient is allergic to it.
   b) Any patient allergic to Antibiotic-A is prescribed Antibiotic-B.
   c) Any patient with a high fever and positive culture has bacterial infection.
   d) Any patient whose skin test is positive is allergic to Antibiotic-A.
   e) Patient Liza has high fever.
   f) Patient Liza has positive culture.
   g) Patient Liza has positive skin test.

h) Patient Omar has high fever.
i) Patient Omar has positive culture.
j) Patient Omar does not have positive skin test.
k) Patient Neha has high fever.
l) Patient Neha does not have positive culture.

Determine which medicine, if any, should be prescribed to Liza, Omar, and Neha. Show the logical derivation clearly.

4. Consider the following axioms:
   a) Any student who has completed Calculus-I may register for Calculus-II.
   b) Any student who has unpaid tuition may not register for any course.
   c) Anyone who received full waiver has no unpaid tuition.
   d) Any student who may register for Calculus-II and has completed Physics-I may register for Engineering Mechanics.
   e) Samin has completed Calculus-I.
   f) Samin has completed Physics-I.
   g) Samin has unpaid tuition.
   h) Arpa has completed Calculus-I.
   i) Arpa has completed Physics-I.
   j) Arpa received full waiver.
   k) Tonmoy has completed Calculus-I.
   l) Tonmoy has no record of completing Physics-I.

Determine whether Samin, Arpa, and Tonmoy may register for Calculus-II and Engineering Mechanics. Show the logical derivation clearly.